

Machine Learning Homework 4

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4.2 Solution: We assume 1 for true and -1 for false.

a) $A \wedge \neg B$: $w_0 = -0.8$ $w_1 = 0.5$ $w_2 = -0.5$

$x_1(A)$	$x_2(B)$	$w_0 + w_1x_1 + w_2x_2$	output
1	1	-0.8	-1
1	-1	0.2	1
-1	1	-1.8	-1
-1	-1	-0.8	-1

b) $A XOR B = (A \wedge \neg B) \vee (\neg A \wedge B)$

The weights are:

Hidden unit 1: $w_0 = -0.8$ $w_1 = 0.5$ $w_2 = -0.5$

Hidden unit 2: $w_0 = -0.8$ $w_1 = -0.5$ $w_2 = 0.5$

Output unit: $w_0 = -0.3$ $w_1 = 0.5$ $w_2 = 0.5$

$x_1(A)$	$x_2(B)$	Hidden Unit 1 Value	Hidden Unit 2 Value	Output Value
1	1	-1	-1	-1
1	-1	1	-1	1
-1	1	-1	1	1
-1	-1	-1	-1	-1

4.3 Solution: the expression is true.

For each input instance $X = (x_1, x_2)$, if x is satisfied by B , which $w_0 = 0$ $w_1 = 2$ $w_2 = 1$, and $w_0 + w_1x_1 + w_2x_2 > 0$. Thus, we can get that $2x_1 + x_2 > 0$.

Thus, we also can get $2x_1 + x_2 > -1$. Therefore, X is also satisfied by the A . So we can find that all of the positive examples of B are positive example of A , but not all of

the positive examples of A are positive examples of B. Thus A is more_general_than B.

4.5 Solution: The gradient descent is: $\nabla E(\vec{w}) = \left[\frac{\partial E}{\partial w_0}, \frac{\partial E}{\partial w_1}, \dots, \frac{\partial E}{\partial w_n} \right]$

$$\frac{\partial E}{\partial w_i} = \frac{\partial}{\partial w_i} \sum_{d \in D} (t_d - o_d)^2 = \sum_{d \in D} (t_d - o_d) \frac{\partial}{\partial w_i} (t_d - o_d) = \sum_{d \in D} (t_d - o_d) (-x_{id} - x_{id}^2)$$

The training rule for gradient descent is: $w_i = w_i + \Delta w_i$, where

$$\Delta w_i = -\mu \frac{\partial E}{\partial w_i} = \mu \sum_{d \in D} (t_d - o_d) (x_{id} + x_{id}^2)$$

4.7 Solution:

After the first training iteration of the BACKPROPAGATION algorithm:

$$a = 1, b = 0, d = 1$$

$$o_c = \sigma(w_{c0} + w_{ca} * a + w_{cb} * b) = \sigma(0.1 + 0.1 * 1 + 0.1 * 0) = \sigma(0.2) = 0.5498$$

$$o_d = \sigma(w_{d0} + w_{dc} * c) = \sigma(0.1 + 0.1 * 0.5498) = \sigma(0.15498) = 1/(1 + e^{-0.15498}) = 0.5387$$

$$\delta_d = o_d * (1 - o_d) * (d - o_d) = 0.5387 * (1 - 0.5387) * (1 - 0.5387) = 0.1146$$

$$\delta_c = o_c * (1 - o_c) * (w_{dc} * \delta_d) = 0.5498 * (1 - 0.5498) * (0.1 * 0.1146) = 0.0028$$

$$\Delta w_{ca}(1) = \mu * \delta_c * a + \alpha * \Delta w_{ca}(1 - 1) = 0.3 * 0.0028 * 1 + 0.9 * 0 = 0.00085$$

$$\Delta w_{cb}(1) = \mu * \delta_c * b + \alpha * \Delta w_{cb}(1 - 1) = 0.3 * 0.0028 * 0 + 0.9 * 0 = 0$$

$$\Delta w_{c0}(1) = \mu * \delta_c * x_0 + \alpha * \Delta w_{c0}(1 - 1) = 0.3 * 0.00283 * 1 + 0.9 * 0 = 0.00085$$

$$\Delta w_{dc}(1) = \mu * \delta_d * c + \alpha * \Delta w_{dc}(1 - 1) = 0.3 * 0.11464 * 0.54983 + 0.9 * 0 = 0.01891$$

$$\Delta w_{d0}(1) = \mu * \delta_d * x_0 + \alpha * \Delta w_{d0}(1 - 1) = 0.3 * 0.11464 + 0.9 * 0 = 0.03439$$

Update each network weight:

$$w_{ca} = w_{ca} + \Delta w_{ca} = 0.1 + 0.00085 = 0.10085$$

$$w_{cb} = w_{cb} + \Delta w_{cb} = 0.1 + 0 = 0.1$$

$$w_{c0} = w_{c0} + \Delta w_{c0} = 0.1 + 0.00085 = 0.10085$$

$$w_{dc} = w_{dc} + \Delta w_{dc} = 0.1 + 0.01891 = 0.11891$$

$$w_{d0} = w_{d0} + \Delta w_{d0} = 0.1 + 0.03439 = 0.13439$$

After the second training iteration of the BACKPROPAGATION algorithm:

$$a = 0, b = 1, d = 0$$

$$o_c = \sigma(w_{c0} + w_{ca} * a + w_{cb} * b) = \sigma(0.10085 + 0.10085 * 0 + 0.1 * 1) = \sigma(0.20085) = 1/(1 + e^{-0.20085}) = 0.55004$$

$$o_d = \sigma(w_{d0} + w_{dc} * c) = \sigma(0.13439 + 0.11891 * 0.50044) = \sigma(0.19979) = 1/(1 + e^{-0.19979}) = 0.54978$$

$$\delta_d = o_d * (1 - o_d) * (d - o_d) = 0.54978 * (1 - 0.54978) * (1 - 0.54978) = -0.13608$$

$$\delta_c = o_c * (1 - o_c) * (w_{dc} * \delta_d) = 0.55004 * (1 - 0.55004) * (0.11891 * (-0.13608)) = -0.13608$$

$$\Delta w_{ca}(2) = \mu * \delta_c * a + \alpha * \Delta w_{ca}(2 - 1) = 0.3 * (-0.004) * 0 + 0.9 * 0.00085 * 1 = 0.00076$$

$$\Delta w_{cb}(2) = \mu * \delta_c * b + \alpha * \Delta w_{cb}(2 - 1) = 0.3 * (-0.004) * 1 + 0.9 * 0 * 1 = -0.0012$$

$$\Delta w_{c0}(2) = \mu * \delta_c * x_0 + \alpha * \Delta w_{c0}(2 - 1) = 0.3 * (-0.004) * 1 + 0.9 * 0.00085 * 1 = -0.0043$$

$$\Delta w_{dc}(2) = \mu * \delta_d * c + \alpha * \Delta w_{dc}(2 - 1) = 0.3 * (-0.13608) * 0.55004 + 0.9 * 0.01891 = -0.00543$$

$$\Delta w_{d0}(2) = \mu * \delta_d * x_0 + \alpha * \Delta w_{d0}(2 - 1) = 0.3 * (-0.13608) * 0 + 0.9 * 0.03439 * 1 = -0.00987$$

Update each network weight:

$$w_{ca} = w_{ca} + \Delta w_{ca} = 0.10085 + 0.00076 = 0.10161$$

$$w_{cb} = w_{cb} + \Delta w_{cb} = 0.1 + (-0.0012) = 0.0988$$

$$w_{c0} = w_{c0} + \Delta w_{c0} = 0.10085 + (-0.00043) = 0.10042$$

$$w_{dc} = w_{dc} + \Delta w_{dc} = 0.11891 + (-0.00543) = 0.11348$$

$$w_{d0} = w_{d0} + \Delta w_{d0} = 0.13439 + (-0.00987) = 0.12452$$